OR4023 (Optimization) Homework 1

Name:_____ Student ID:_____

Question 1. Prove the open ball

$$B_R = \{y \in \mathbb{R}^n : \|y\| < R\}$$

is open and convex.

Question 2. Prove the intersection of two convex sets is convex.

Question 3. Find the first and second order Taylor series expansion of the function

$$f(x, y) = xe^y + \sin(xy)$$

at $(0, 0)$ and estimate the error for the first order approximation.

Question 4. Prove that if $f(x)$ and $g(x)$ are two convex functions defined on a convex set $S \subseteq \mathbb{R}^n$, then $f(x) + g(x)$ is also a convex function on S.

Question 5. Consider the function $f : \mathbb{R} \to \mathbb{R}$ defined piecewise as follows:

$$f(x) = \begin{cases} 0 & \text{if } x \leq 0, \\ x^2 & \text{if } x > 0. \end{cases}$$

a) Show that f is convex by definition.

b) Show that f is differentiable everywhere.

c) Show that f is convex by the first derivative test.

Question 6. Let $f(x_1, x_2) = x_1^2 - x_1x_2 + x_2^2$. Is f convex on $\mathbb{R}^2$?

Question 7. Fit the data points $(0, 1), (1, 3), (2, 2)$ by a linear function $y(t) = x_1t + x_2$ using the least squares method with the following steps.

a) Formulate an optimization problem.

b) Find the stationary point and show it is a global minimizer.

c) Plot a graph to illustrate your result.